

Lecture 4

semi convergence

of the MR iterative algorithm

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Appetizer... Eigenvalue decomposition

- K is a regular $m \times m$ matrix
- $K\phi_i = \mu_i\phi_i : \mu_i \in \mathbb{R}, \phi_i \in \mathbb{R}^m$ is K 's eigen system
- Let p be a polynomial : (e.g. $p(K) = I + K + 2K^2$)
- $\mathbf{e} \in \mathbb{R}^m$

How to bound $\|\mathbf{e}_k\| := \|p(K)\mathbf{e}\|$?

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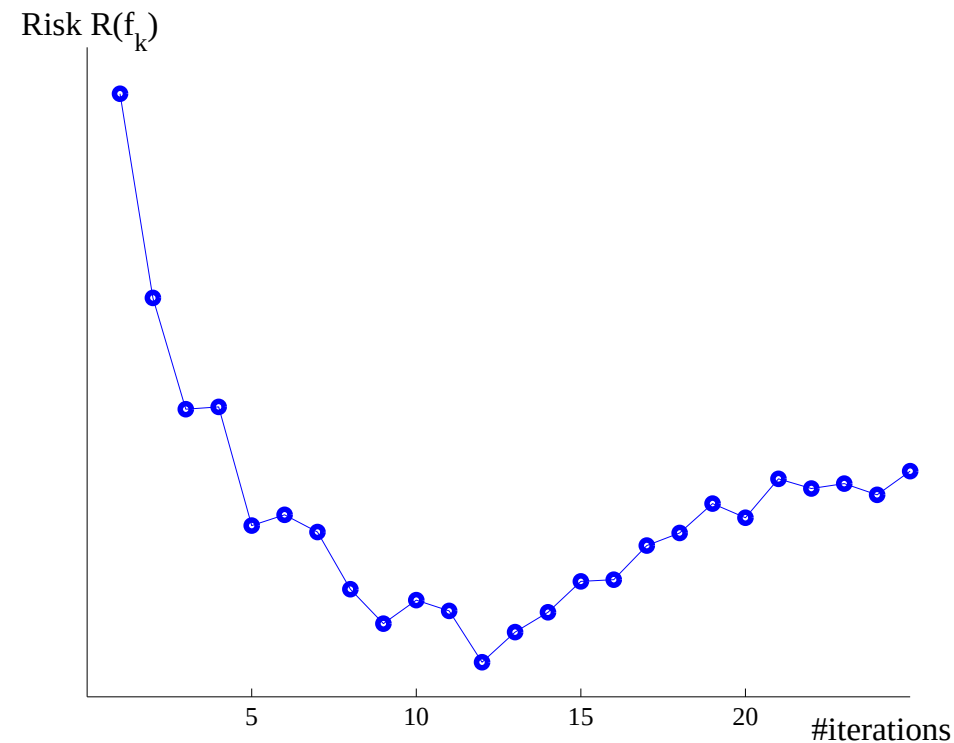
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$(\mu^* := \operatorname{argmax}_{i=1, m} |p(\mu_i)|)$

Road map



- the minimal residual algorithm
- semi convergence
 - the theorem
 - proof strategy
 - from risk to vector
 - "small" eigen values
 - "large" eigen values
- bound of the empirical cost
 - PAC result

there exists an increasing sequence $p'_k(0)$ such that

$$\sqrt{R(f_k) - R(t)} \leq \frac{a}{|p'_k(0)|} + b|p'_k(0)| + c$$

Efficient Krylov subspace algorithm

$$K\alpha = \mathbf{y} \quad \min_{\alpha \in \mathbf{R}^m} \|K\alpha - \mathbf{y}\|^2$$

$$\alpha = \sum_{i=1}^m a_i \mathbf{d}_i \quad \text{with} \quad \mathbf{d}_i^\top K K \mathbf{d}_j = 0, i \neq j$$

The minimum residual algorithm

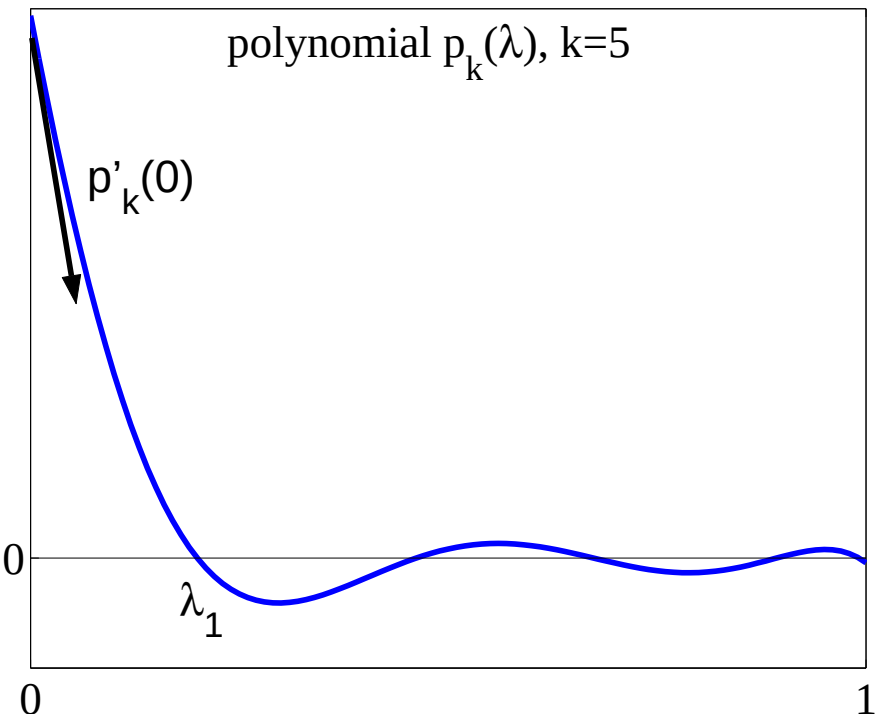
$$\begin{cases} (1) & \alpha_k = \alpha_{k-1} + a_k \mathbf{d}_k \\ (2) & \mathbf{r}_k = \mathbf{y} - K\alpha_k \\ (3) & \mathbf{d}_{k+1} = \mathbf{r}_k + b_k \mathbf{d}_k \end{cases}$$

The polynomial interpretation

$$\alpha_k = q_{k-1}(K)\mathbf{y} \quad r_k = p_k(K)\mathbf{y}$$

$$r_k = \mathbf{y} - K\alpha_k \implies p_k(K) = I - Kq_{k-1}(K)$$

$$\alpha_k \in \text{span}\{\mathbf{y}, K\mathbf{y}, K^2\mathbf{y}, \dots, K^{k-1}\mathbf{y}\}$$



Semi-convergence result

Framework

■ $t(x) = \mathbb{E}(Y|X = x)$ is the target function, $R(t) = \mathbb{E}((t(X) - Y)^2)$

■ $\mathcal{H}$ is a R.K.H.S. with kernel κ , find in $\mathcal{H}$ a regularized solution of :

$$Tf = y \Leftrightarrow K\alpha = y \quad K_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$$

■ $Tt = \mathbf{y}_t, K\alpha_t = \mathbf{y}_t$

■ use MR algorithm to generate a sequence $\alpha_k = q_{k-1}(K)y$, with

$$p_k(K) = I - Kq_{k-1}(K)$$

■ back to the solution : $f_k(.) = T^*\alpha_k = \sum_{i=1}^m \alpha_{ki} \kappa(x_i, .)$

Theorem (semiconvergence of the MR algorithm)

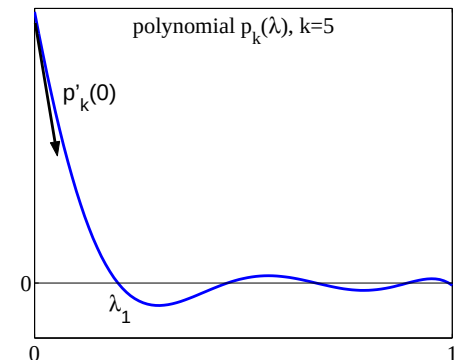
If there exists some vector ω such that $\alpha_t = K\omega$, (if the target function is regular) then

$$\sqrt{R(f_k) - R(t)} \leq \frac{a}{|p'_k(0)|} + b|p'_k(0)| + c$$

■ $a = \|T^*\| \|\omega\|$

■ $b = 3\|T^*\| R_{\text{emp}}(t)$

■ $c = 2\|\alpha_t\| + \|t - t^*\|$



Proof strategy $\sqrt{R(f_k) - R(t)} \leq \frac{a}{|p'_k(0)|} + b|p'_k(0)| + c$

1 from the risk $R(f_k)$ to the vector α_k with $T^* \alpha_t = t^*$

$$\sqrt{R(f_k) - R(t)} \leq \|f_k - t\| \leq a' \|\alpha_k - \alpha_t\| + b'$$

2 the cutting trick

split $\|\alpha_k - \alpha_t\|$ in two parts with **small** and **large** eigen values

3 control the small eigen values

CRITIC part

4 control the large eigen values term by

$$\frac{1}{\varepsilon} R_{\text{emp}}(f_k)$$

5 control $R_{\text{emp}}(f_k)$:

BONUS - PAC bounds because $\mathcal{H}$ is Glivenko-Cantelli

Prove that : $\sqrt{R(f_k) - R(t)} \leq \frac{a}{|p'_k(0)|} + b|p'_k(0)| + c$

$$\begin{aligned} R(f_k) - R(t) &= \mathbb{E}(f_k(X) - Y)^2 - \mathbb{E}(t(X) - Y)^2 \\ &= \mathbb{E}(f_k(X)^2 - 2f_k(X)Y + Y^2 - t(X)^2 + 2t(X)Y - Y^2) \\ &= \mathbb{E}(f_k(X)^2 - 2f_k(X)t(X) - t(X)^2 + 2t(X)t(X)) \\ &= \mathbb{E}(f_k(X)^2 - 2f_k(X)t(X) + t(X)^2) = \|f_k - t\|^2 \end{aligned}$$

let define t^* the projection of t onto the domaine of T^* , i.e. $\exists \alpha_t \in \mathbb{R}^n$ such that $t^* = T^* \alpha_t$ and $Tt^* = Tt = y_t$.

$$\begin{aligned} \|f_k - t\| &\leq \|f_k - t^*\| + \|t^* - t\| \\ &\leq \|T^* \alpha_k - T^* \alpha_t\| + \|t^* - t\| \\ &\leq \|T^*\| \|\alpha_k - \alpha_t\| + \|t^* - t\| \end{aligned}$$

we have

$$\sqrt{R(f_k) - R(t)} \leq \|T^*\| \underbrace{\|\alpha_k - \alpha_t\|}_{\text{}} + \|t^* - t\|$$

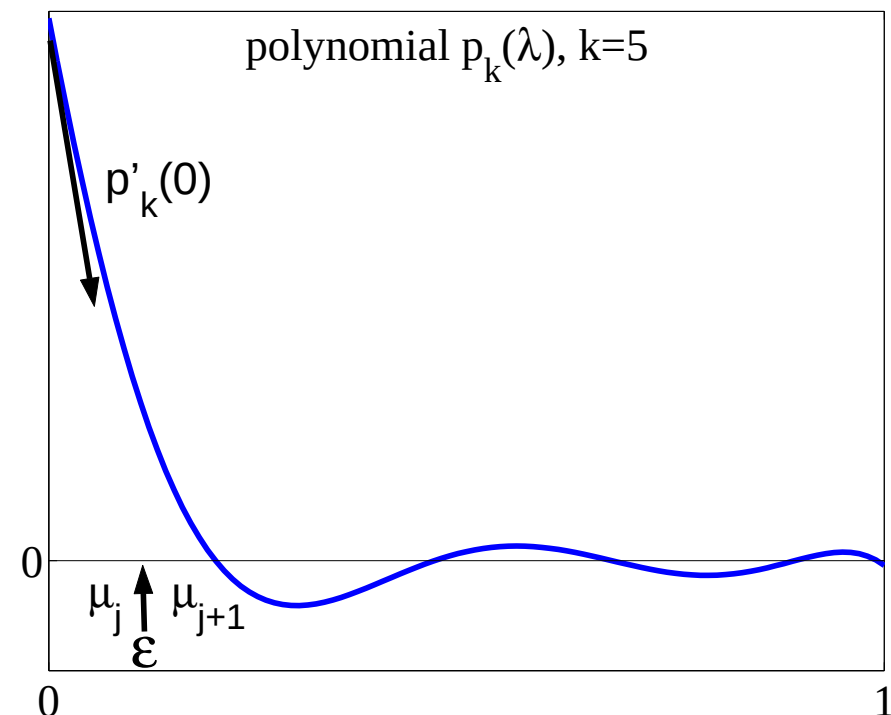
The cutting trick

$K\phi_i = \mu_i\phi_i$, assume $0 < \mu_1 < \dots < \mu_m \leq 1$ and $\|\phi_i\|^2 = 1$,

$$\forall \mathbf{e} \in \mathbb{R}^m, \|\mathbf{e}\|^2 = \|\mathbf{e}\|^2 = \left\| \sum_{i=1}^m (\mathbf{e}^\top \phi_i) \phi_i \right\|^2 = \sum_{i=1}^m (\mathbf{e}^\top \phi_i)^2$$

Let take an ε such that $0 < \varepsilon \leq \frac{1}{|p'_k(0)|}$. Assume $\mu_j \leq \varepsilon < \mu_{j+1}$

$$\|\alpha_k - \alpha_t\| \leq \left(\sum_{i=1}^j \underbrace{((\alpha_k - \alpha_t)^\top \phi_i)^2}_{\text{small eigenvalues}} \right)^{\frac{1}{2}} + \left(\sum_{i=j+1}^n \underbrace{((\alpha_k - \alpha_t)^\top \phi_i)^2}_{\text{large eigenvalues}} \right)^{\frac{1}{2}}$$



Bounding the "small" eigenvalues

ε is such that $0 < \varepsilon \leq \frac{1}{|p'_k(0)|}$. By hypothesis $K\omega = \alpha_t$

Let define $\tilde{\alpha}_t = q_{k-1}(K)y_t$ (recall that $\alpha_k = q_{k-1}(K)y$)

$\alpha_k - \tilde{\alpha}_t = q_{k-1}(K)y - q_{k-1}(K)y_t$ and $\tilde{\alpha}_t - \alpha_t = q_{k-1}(K)y_t - \alpha_t = q_{k-1}(K)K\alpha_t - \alpha_t$

$$\begin{aligned}
 \sqrt{\sum_{i=1}^j ((\alpha_k - \alpha_t)^\top \phi_i)^2} &= \left(\sum_{i=1}^j ((\alpha_k - \tilde{\alpha}_t + \tilde{\alpha}_t - \alpha_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} \\
 &\leq \left(\sum_{i=1}^j ((q_{k-1}(K)(y - y_t))^\top \phi_i)^2 \right)^{\frac{1}{2}} + \left(\sum_{i=1}^j (p_k(K)\alpha_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} \\
 &\leq q_k(\mu^*) \left(\sum_{i=1}^j ((y - y_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} + \left(\sum_{i=1}^j (p_k(K)K\omega)^\top \phi_i)^2 \right)^{\frac{1}{2}} \\
 &\leq |p'_k(0)| \left(\sum_{i=1}^j ((y - y_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} + \varepsilon \left(\sum_{i=1}^j (\omega^\top \phi_i)^2 \right)^{\frac{1}{2}} \\
 &\leq |p'_k(0)| R_{\text{emp}}(t) + \frac{1}{|p'_k(0)|} \|\omega\|
 \end{aligned}$$

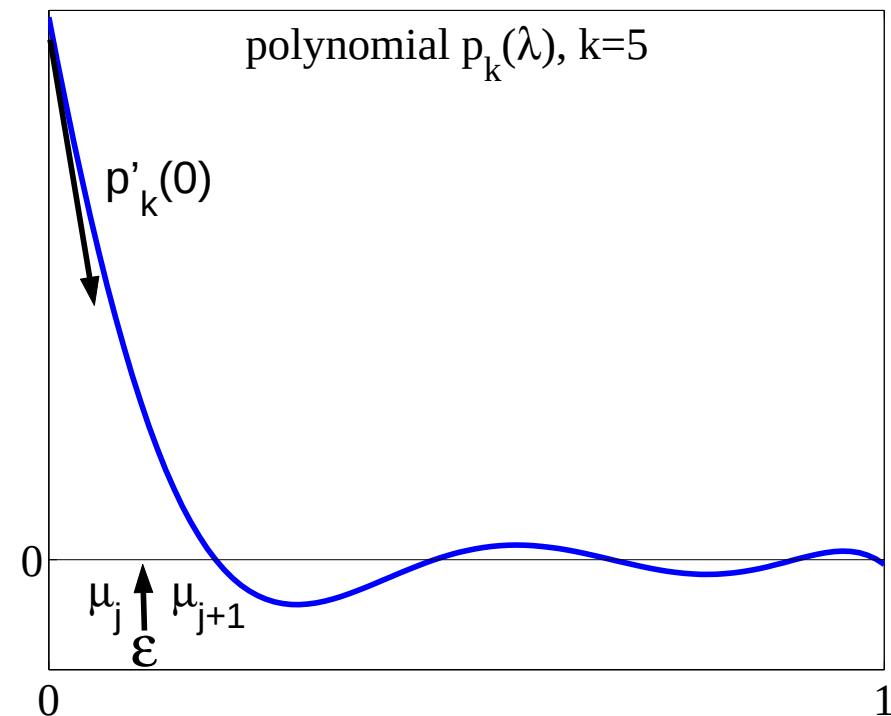
tedious technical points

$$\forall \lambda \in [0, \varepsilon], \lambda \leq \varepsilon \Rightarrow p_k(\lambda)\lambda \leq \varepsilon \text{ since } p_k(\lambda) \leq 1$$

and

since $\varepsilon < \lambda_{1k}$, $p_k(\lambda)$ is convex in the interval $[0, \varepsilon]$ and $p_k(0) = 0$

$$\forall \lambda \in [0, \varepsilon], \quad q_k(\lambda) = \frac{1 - p_k(\lambda)}{\lambda} \leq |p'_k(0)|$$



$$\alpha_k = q_{k-1}(K)\mathbf{y}$$

$$r_k = p_k(K)\mathbf{y}$$

$$r_k = \mathbf{y} - K\alpha_k$$

$$\Rightarrow p_k(K) = I - Kq_{k-1}(K)$$

$$\Rightarrow p_k(\mu) = 1 - \mu q_{k-1}(\mu)$$

Bounding the "large" eigenvalues

- the solution $K\alpha = \mathbf{y}$ and the computed sequence $K\alpha_k = \mathbf{y}_k$
-

$$\begin{aligned}\sqrt{\sum_{i=j+1}^n ((\alpha_k - \alpha_t)^\top \phi_i)^2} &= \left(\sum_{i=j+1}^n ((K^{-1}\mathbf{y}_k - K^{-1}\mathbf{y}_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} \\ &= \left(\sum_{i=j+1}^n ((\mathbf{y}_k - \mathbf{y}_t)^\top K^{-1} \phi_i)^2 \right)^{\frac{1}{2}} \\ &\leq \frac{1}{\varepsilon} \left(\sum_{i=j+1}^n ((\mathbf{y}_k - \mathbf{y}_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} \\ &\leq \frac{1}{\varepsilon} \left(\sum_{i=j+1}^n ((\mathbf{y}_k - \mathbf{y})^\top \phi_i)^2 \right)^{\frac{1}{2}} + \frac{1}{\varepsilon} \left(\sum_{i=j+1}^n ((\mathbf{y} - \mathbf{y}_t)^\top \phi_i)^2 \right)^{\frac{1}{2}} \\ &\leq \frac{1}{\varepsilon} (R_{\text{emp}}(f_k) + R_{\text{emp}}(t))\end{aligned}$$

the empirical risk has to be controlled as well

Controlling the empirical risk

Theorem (Convergence rate of the empirical cost for the minimal residual error Krylov space iterative algorithm)

Let f_k be the sequence of functions given by the MR algorithm p_k the associated residual polynomial t the target function and

if $\exists t^* \in \mathcal{H}$ such that $Tt^* = y_t$ and $K\alpha_t = y_t$

then

$$R_{\text{emp}}(f_k) \leq R_{\text{emp}}(t) + \frac{2}{|p'_k(0)|} \|\alpha_t\|$$

proof

$$\begin{aligned} R_{\text{emp}}(f_k) &= \|y - K\alpha_k\| \\ &= \|p_k(K)y\| \\ &= \|p_k(K)(y - y_t + y_t)\| \\ &\leq \|p_k(K)(y - y_t)\| + \|p_k(K)y_t\| \\ &\leq \|p_k(K)\| \|y - y_t\| + \|p_k(K)K\alpha_t\| \leq R_{\text{emp}}(t) + \frac{2}{|p'_k(0)|} \|\alpha_t\| \end{aligned}$$

Needs to prove that $\|p_k(K)\| \leq 1$ and $\|p_k(K)K\| \leq \frac{2}{|p'_k(0)|}$

Conclusion

- learning is ill posed

- regularize !

- choose pointwise functions $\mathcal{H} \in \mathbb{R}^{\Omega}$
- transform function to vectors
- use one regularization mechanism
 - penalization
 - subspace selection
 - iterative method

- use iterative algorithm early stopping to regularize

- it's fast
- it's generic
- it's accurate : good empirical results
- there is semiconvergence proof